Jacopo Zacchigna

MSc Student · Mechanistic Interpretability · AI Safety · London, UK

jaczac2002@gmail.com | jac-zac.github.io | Jac-Zac | jacopo-zacchigna | Google Scholar

Research Experience

UCL, *Research Visitor in Computational Linguistics* – London, UK — Oct 2026 – present

White-box auditing of LLM agents with *Mario Giulianelli* and *Gabriele Sarti*, in collaboration with Parallx.

- Built a NASim–Inspect cyber environment and a dataset of 8,291 Gemma-4-12B agent decisions across 348 worlds.
- Probing activations for hidden network state and next actions on held-out worlds, vs. history-only baselines.

Supervised Program for Alignment Research (SPAR), *Research Fellow* – Remote — Feb 2026 – May 2026

With *Gabriele Sarti* on how LLMs implicitly represent their users (report, code).

- Released persona vectors (Gemma-3-27B, Llama-3.1-70B/405B) and a web interface for probing and steering.
- Found a decodability–control gap: user attributes were linearly decodable from Gemma-3-27B activations (e.g. 97% on 10-way residence), yet ActAdd steering shifted answers by at most 3 points.

Selected Projects

Expert Specialization in Mixture-of-Experts LLMs, code + report — Jan 2026 – Aug 2026

- Decomposed OLMoE expert outputs into sparse token directions via Head Pursuit; replicated on GPT-OSS-20B.
- Knocking out the top 1% of experts cut country-token propensity by ~21% but an offensive-word probe by only ~5%.

Causal Self-Attention Kernels, code + report — Jan 2026 – Feb 2026

- Wrote C++ (SIMD, OpenMP) and CUDA attention kernels from naive up to Flash-Attention-style online softmax.
- OpenMP kernel: 94× speedup on 128 cores (73% efficiency), within 1.5× of PyTorch's fused SDPA in wall-clock and out-scaling it (73% vs 50% parallel efficiency); CUDA kernel 1.09× faster than naive PyTorch on A100.

Publications

Zacchigna, J. & Liu, W. (co-first authors), et al. *Application of Computer Vision to the Automated Extraction of Metadata from Natural History Specimen Labels: A Case Study on Herbarium Specimens.* Plants 15(4) — 2026

Grants & Awards

BlueDot Impact Rapid Grant ($6,000) for white-box auditing of LLM agents' beliefs — 2026

Aletheia's Quest LLM lie-detection competition: 3rd place, black-box track — 2026

Education

University of Trieste, MSc, Data Science & Artificial Intelligence — 2024 – 2027 (expected)

University of Trieste, BSc, AI & Data Analytics — 2021 – 2024

Industry Experience

Obloo, *AI Consultant* – Trieste, Italy — May 2024 – Jan 2025

- Designed and deployed retrieval-augmented generation (RAG) systems over the company data lake.
- Mentored 5 interns to independent project delivery; led ML prototypes adopted in two company products.

Skills

Technical: Python, PyTorch, nnsight, Hugging Face, Inspect, C/C++, CUDA, OpenMP, MPI, Slurm, Docker, Linux
Languages: Italian (native), English (C1)